Physics 7502 **Instructor** **Prof. Justin Frantz**

Particles and Nuclei 2 : - Particle Physics and QCD

Spring 2026

Office: Edwards Accelerator Lab 205 **e-mail:** frantz@ohio.edu
Meeting Times: Tu/Th 1:30-2:50
Location: Clippinger 389
Office hours: For now as needed, by appointment, most likely Wednesdays ~2pm.

Website: Canvas. canvas.ohio.edu This syllabus, homework, and additional information about the course can be found there. THIS SYLLABUS IS SUBJECT TO CHANGE DURING THE QUARTER: Please check the canvas page for the most current version. However, I will notify you of any important changes.

Textbooks (required):
Introduction to Elementary Particles 2nd Edition, David Griffiths [Much of this book will be covered]

An Introduction to Particle Physics and the Standard Model, (Free Online –Open Access) Robert Mann [A few chapters]

Alt Textbooks (optional) *strongly recommended—will refer to a good bit

*Quantum Field Theory for the Gifted Amateur Lancaster, Tom Lancaster and Stephen Blundell

*Modern Particle Physics Thomson, Mark ISBN 10: 1107034264 (Another good one) (ebook available ~$50)

Quarks and leptons: an Introductory course in Modern Particle Physics, Halzen, Francis and Martin, Alan D. (Free online?)

Grading (Approximate, to be revised/confirmed in class)
Problem sets: 55%
Midterm Exam & Final Exam/Presentation/Project: 35% (All Exams will be "Take-Home" Exams)
Participation, Quizzes, Special Projects: 10%

Within these approximate guidelines, I may take some weight from certain portions of the grade and apply it towards others, if it is to the students advantage. For example if the midterm performance is especially good, it may count more of the final grade, while then the final exam may instead count for less. Such changes will be limited to +- 10% in each category. Special projects may include some in-class presentations for example.

In-class exercises and activities will be a part of the classes, in some cases, possibly many, these will be collected, missing an excessive amount of class or excessively *not* participating ~at all, could cause your grade to be lowered.

Homework & Problem Sets
Homework sets or other kinds of homework (e.g. reviewing and presenting some short paper topics) will be assigned about once every week and a half to two weeks on average. Typically there are 7 to 8 HW assignments, (in addition to a take home midterm and final/project/paper).

Readings: The readings will be outlined in class on the first slides (on canvas) of every class.

Preliminary Course Content And Approximate Schedule

-Weeks 1-4 : Griffiths Chapters 1,2,4,5 –Related Material (Basics, Symmetries)
-Weeks 3-6 : Griffiths Ch 3, Kinematics / Rapidity ! Detectors/Accelerators Mann Chapters - Possibly some topics
-Weeks 6-11 - ish Griffiths Chapters 6-8 (9) Related Material (Dirac Equation QED QCD)
-Weeks 11-14 Special Topics Including, More Detector/Accelerators, Jets, Neutrinos, Rel. Heavy Ion, Jlab, and EIC Physics Including Griffiths Chapters (9) 10-12

Learning Outcomes
After this course students will be able to:

Explain the basis of Feynman techniques used to calculate cross-sections for various processes as well as decay width and lifetime of particle resonances,
- Distinguish the three fundamental interactions felt by particles and connect them to results from modern particle physics experiment,
- Describe the theory of the Standard model of Particle Physics.
- Describe the range of Medium and High Energy QCD Nuclear Physics topics
- Describe key parts of the experimental detectors, accelerator/colliders and methods for experimental measurements related to the above topics.

Attendance & Absence Policy
Attendance in lectures is expected, but a couple times missed shouldn't affect your grade, see previous. If you will miss a substantial number of classes, schedule a discussion with me.

Academic Misconduct Policy
I remind you that academic misconduct is a Code A violation of the Ohio University Code of Student Conduct. If you are found to be involved in academic misconduct regarding this course, you will receive F on the pertinent work and possibly for the entire course and/or a referral to the Director of Judiciaries. Procedures for judicial actions may be invoked as described in the Student and Faculty Handbooks. One thing to be careful of is blatently copying solutions you find on the web, especially without fully understanding them.

Reasonable Religious Accomodations:
In addition to participation/attendance/absence policies already listed, students may be absent for up to three (3) days each academic semester, without penalty, to take time off for reasons of faith or religious or spiritual belief system or to participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization. Students/You are required to notify the instructor/me in writing of specific dates requested for alternative accommodations no later than fourteen (14) days after the first day of instruction. These requests will remain confidential. For more information about this policy, students/you may contact the Director and Title IX Coordinator, Equity and Civil Rights Compliance, Lindley Hall, 006, 740-593-9140, Equity@ohio.edu